M541: Stocastic Processes - MCMC Exploration Using Audio

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1 Introduction

It is no surprise to those in the larger study of statistics, that the dominating approach is that of the frequentist. However, there is has been a rise in popularity of the lesser used Bayesian approach, where the need for describing and analyzing complex data has become increasing relevant in fields such as medicine and artificial intelligence. Hence, knowledge of this approach becomes more appropriate for students to gain an understanding of before going into industrial fields or academia. This is where the following exploration of a specific concept in Bayesian statistics arises from. For the Fall semester of 2026, I had the opportunity to take a graduate level course in stochastic processes. Many of the concepts covered in the course range from various Markovian processes, to an introduction into Brownian motion. A particular concept that we were afforded to study on our own was that of Markov Chain Monte Carlo (MCMC). This markup provides a summary of some introductory material that one who might self-study this in an undergraduate degree might come across. The postceding sections will provide an introduction into the theory of MCMC, and will then delve into an exploration of an implementation of the Metropolis-Hastings algorithm using audio data sampled from a musical piece. The findings will then briefly be discussed.

2 Markov Chain Monte Carlo

Before expressing the formulation of MCMC, a look into the works of Bayes and Ulam are needed in order to understand the evolution into the methods seen in Bayesian statistics in the modern era. That is, these works are the foundational stepping stones that provide a path into what is now the Monte Carlo Method. Generally speaking, Monte Carlo methods were introduced during World War II in the Los Alamos National Laboratory, where these methods would later be used to form the Metropolis algorithm by the early 1950s [2]. Now, jumping backwards some 210 years prior, Bayes' first developed what would later become a published theorem, used in all of Bayesian statistics today; it is stated mathematically as

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}, \tag{1}$$

where A, B are events such that $P(B) \neq 0$. Here, $P(A|B), P(B|A)$ are conditional probabilities, where A and B depend on the occurence of the other. Now, without going any further into the extensions of Bayes' Theorem, it is noted that Equation (1) was used in order to compute the distribution of an unknown probability parameter θ of a binomial distribution. This is the main contrast from that of the frequentist's perspective; the parameters of a distribution are unknown, and the most likely distribution for those parameters that describe the data are to be found. Thus, Equation (1) is reexpressed as

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}. \tag{2}$$

Equation (2) assumes the parameters of a distribution depend on the data, $P(\theta|D)$ (called the posterior), and this probability is informed by the likelihood $P(D|\theta)$, given a prior assumption,

$P(\theta)$, about the parameters before any new data ($P(D)$) is observed. In other words, MCMC allows for a sampling from any distribution that can not be directly sampled, in essence, sampling from a prior distribution over θ.

2.1 Metropolis-Hastings Algorithm

Now, there are many ways of implementing MCMC; a particular implementation is that of the Metropolis-Hastings algorithm. Essentially, what happens is that given a proposal distribution, denoted $Q(\theta'|\theta)$, helps to draw samples from a posterior distribution $P(\theta|D)$ that is refractory. This works by using Q to perform a random walk within the parameter space, where the transition into a new state is accepted or rejected based on how likely the sample is to be observed. This process is Markovian in nature, hence the "Markov Chain" in MCMC. The likelihood of each new observation is derived from a function f, thus f must be proportional to $P(\theta|D)$. The random walk of Q is formed by a sample drawn from typically from a symmetric distribution proposed by new parameter θ', given the current parameter θ. That is, $Q(\theta'|\theta)$ typically follows some distribution of symmetry such as $Q(\theta'|\theta) = \mathcal{N}(\mu, \theta)$. To decide the acceptance of θ', the ratio

$$\begin{aligned} \frac{P(\theta'|D)}{P(\theta|D)} &= \frac{P(D|\theta')P(\theta')}{P(D|\theta)P(\theta)} \text{(Bayes' Formula)} \\ &= \frac{\prod_i^n f(d_i|\Theta = \theta')P(\theta')}{\prod_i^n f(d_i|\Theta = \theta)P(\theta)} \end{aligned}$$

which leads to the probability that

$$P(\text{accept}) = \begin{cases} \frac{\prod_i^n f(d_i|\Theta=\theta')P(\theta')}{\prod_i^n f(d_i|\Theta=\theta)P(\theta)}, \prod_i^n f(d_i|\Theta = \theta)P(\theta) > \prod_i^n f(d_i|\Theta = \theta')P(\theta') \\ 1, \prod_i^n f(d_i|\Theta = \theta)P(\theta) \leq \prod_i^n f(d_i|\Theta = \theta')P(\theta'). \end{cases} \tag{3}$$

What Equation (2) states is that if θ' is a strong proposal for the next state, it is certain that θ' is chosen. If it is strictly lesser to the current θ, it may still be accepted, however, with decreasing probability. Algorithm 1 provides a psuedocode representation of how this may be implemented [3].

Algorithm 1: Metropolis-Hastings Algorithm

```
Initialise a state θ_0; Set t = 0;
for t to N do
    1. Sample candidate θ' from proposal distribution Q(θ'|θ_t);
    2. Calculate the acceptance ratio:
       α = min(1, P(θ')Q(θ_t|θ') / (P(θ_t)Q(θ'|θ_t)));
    3. Generate u ~ Uniform(0, 1);
    if u ≤ α then
        Accept the proposal: θ_{t+1} ← θ';
    end
    else
        Reject the proposal: θ_{t+1} ← θ_t;
    end
    Increment t = t + 1;
end
```

3 Implementation of MCMC to Audio

The data used in implementation in this study comes from a personal favorite musical artist, that is an artist by the name of Porter Robinson. A sample of piece *Divinity* is taken for study. The sample is at a duration 0.065s at 11.25khz. Figure 1 provides a visual representation of the data used.

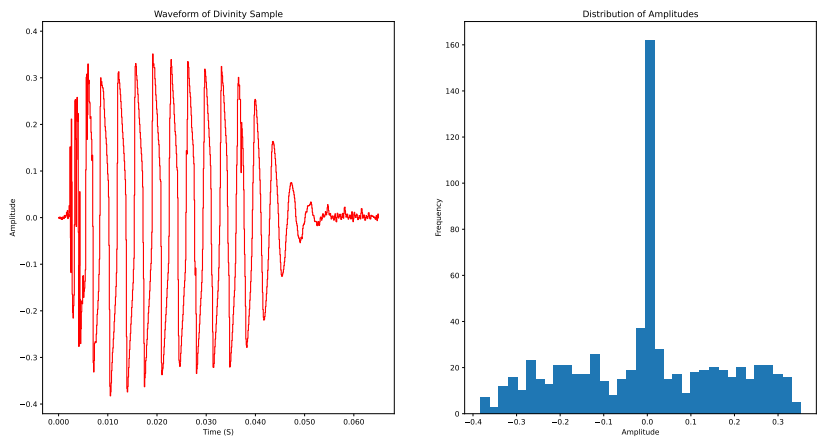

Figure 1: Dual plot of the audio waveform and distribution of amplitudes

As audio generally tends to be classified as sparse data, where the sample representation are zero [1], a distribution with the capacity to model such representations is needed. Other studies have found that a Laplace distribution can be used in order to achieve this [4]. The Laplace PDF is defined by

$$f(x|\mu, b) = \frac{1}{2b} e^{-\frac{|x-\mu|}{b}}, \tag{4}$$

where $\mu \in (-\infty, \infty), b \in [0, \infty)$ are location and scale parmeters, respectively. Using this as the prior distribution of the MCMC, the likelihood function is defined as

$$L(\mu, b|x) = \prod_{i=1}^{n} \frac{1}{2b} e^{-\frac{|x_i-\mu|}{b}} = \left(\frac{1}{2b}\right)^n \exp\left(-\frac{1}{b}\sum_{i=1}^{n} |x_i - \mu|\right). \tag{5}$$

Taking the logarithm of Equation 5, obtained is

$$\ell(\mu, b|x) = -n\ln(2b) - \frac{1}{b}\sum_{i=1}^{n} |x_i - \mu|. \tag{6}$$

This posterior is then used in conjunction with Algorithm 1, in order to sample the distribution of parameters for the data. Further implementation for choice of prior and proposal distribution can be found on the Github Repository associated with this study.

4 Results

The MCMC is observed with $10,000$, $30,000$, and $50,000$ iterations. Table 4 provides a comparison of a single pass of each iterations acceptance rate with the parameter estimates. Figure 2 shows four plots for the pass of 50k iterations, observing the convergence of the first 50 samples, the convergence of all samples taken, as well as the traces for μ, b, respectively. Figure 3 shows a comparison of

Iterations	AR	$\hat{\mu}$	$\hat{b}$
10k	28.22%	0.0020 ± 0.0028	0.1337 ± 0.0045
30k	28.33%	0.0020 ± 0.0027	0.1341 ± 0.0049
50k	28.24%	0.0020 ± 0.0028	0.1344 ± 0.0051

Table 1: Acceptance rate and parameter estimates for μ, b

the distribution of samples from the mean estimates of the paramters for the posterior distribution overlayed on top of the observed data.

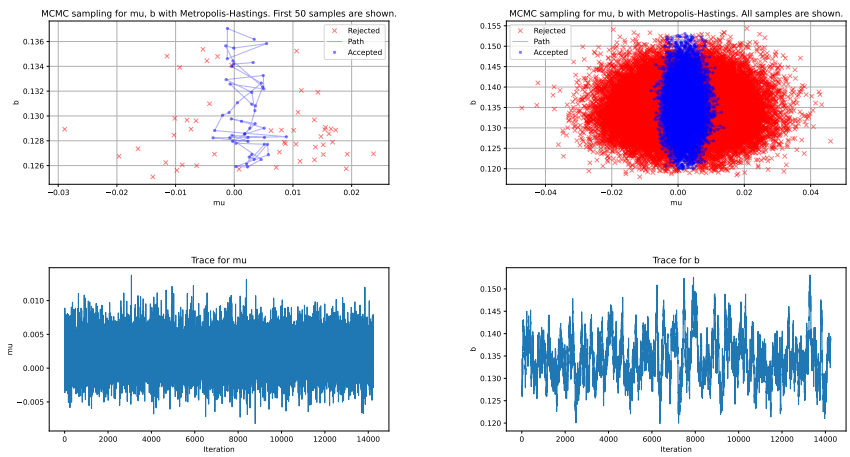

Figure 2: Convergence of parameters, traces of μ, b

5 Discussion

As Table 4 shows, the acceptance rate of a single pass is just shy of what is typically seen in a two-dimensional estimation using MCMC, however, for the nature of the project being an exploration, this is adequate for its purpose. All in all, a brief history and theoretical foundation was touched upon, a look at a particular implementation of MCMC was observed using the Metropolis-Hastings algorithm in the context of estimating the parameter space of a Laplace distribution to sparse audio data. Perhaps in the future, some work towards obtaining a more likely posterior may be attempted, however, for the purposes of self-education, this would seem more was initially set out upon.

6 Thank You

If you, the reader, happened upon this project and read through this entire report, I appreciate your attention. I hope that you learned something new and interesting from this. Thank you.

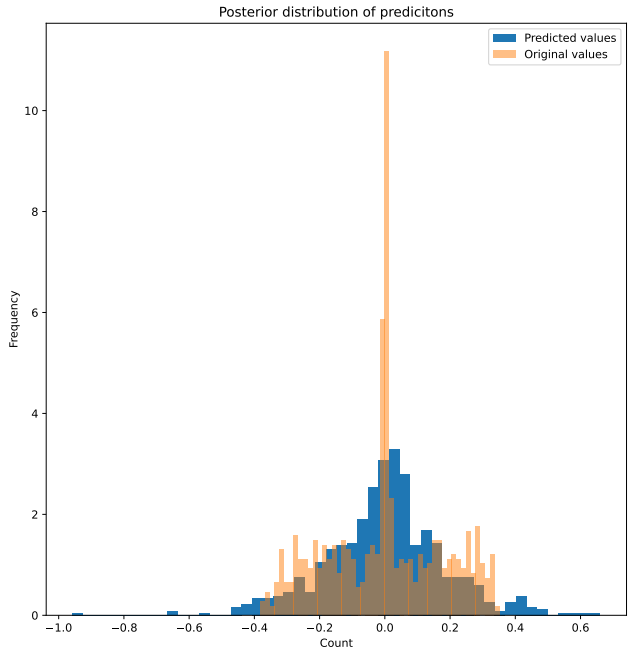

Figure 3: Histogram of original samples and distribution of posterior

References

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[3] Christian P. Robert and George Casella. *Monte Carlo Statistical Methods*. 2nd ed. New York, NY, USA: Springer, 2004. DOI: 10.1007/978-1-4757-4145-2. URL: https://link.springer.com/book/10.1007/978-1-4757-4145-2.

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